

Fig. 3 | Completed model-class background-audit matrix

Background sensitivity is consistent across model architectures.

Target / site	CNN/Mega	LGBM multi	LGBM single	Weis LR
E. coli / Cipro / A-2018	0.82 -> 0.70 ret 62%; S=9	0.83 -> 0.64 ret 62%; S=9	0.83 -> 0.65 ret 62%; S=9	0.73 -> 0.69 ret 61%; S=6
E. coli / Cipro / DRIAMS-C	0.75 -> 0.65 ret 48%; S=4	0.81 -> 0.64 ret 48%; S=4	0.80 -> 0.68 ret 48%; S=4	0.63 -> 0.82 ret 44%; S=2*
E. coli / Cipro / DRIAMS-D	0.67 -> 0.60 ret 98%; S=10	0.72 -> 0.58 ret 98%; S=10	0.71 -> 0.59 ret 98%; S=10	0.67 -> 0.62 ret 91%; S=3
E. coli / Amox-Clav / A-2018	0.65 -> 0.54 ret 92%; S=14	0.63 -> 0.48 ret 92%; S=14	0.61 -> 0.47 ret 92%; S=14	0.56 -> 0.53 ret 94%; S=8
E. coli / Amox-Clav / DRIAMS-C	0.54 -> 0.50 ret 89%; S=10	0.59 -> 0.52 ret 89%; S=10	0.56 -> 0.53 ret 89%; S=10	0.63 -> 0.58 ret 75%; S=3
E. coli / Amox-Clav / DRIAMS-D	0.56 -> 0.49 ret 97%; S=11	0.58 -> 0.45 ret 97%; S=11	0.56 -> 0.49 ret 97%; S=11	0.65 -> 0.52 ret 92%; S=3
S. aureus / Oxa / A-2018	0.84 -> 0.76 ret 60%; S=9	0.79 -> 0.69 ret 60%; S=9	0.84 -> 0.73 ret 60%; S=9	not run
S. aureus / Oxa / DRIAMS-B	0.78 -> 0.86 ret 53%; S=1	0.77 -> 0.81 ret 53%; S=1	0.82 -> 0.87 ret 53%; S=1	0.67 -> 0.67 ret 100%; S=1*
S. aureus / Oxa / DRIAMS-C	0.73 -> 0.70 ret 60%; S=2	0.66 -> 0.62 ret 60%; S=2	0.72 -> 0.73 ret 60%; S=2	0.54 -> 0.54 ret 100%; S=1

Cells show raw AUC followed by background-centered AUC for CNN/Mega, multi-task LightGBM, single-task LightGBM and Weis/Borgwardt LR rows. CNN and LightGBM use threshold matched support; N/A = no usable S/R oxacillin-labeled *S. aureus* expanded study panels. Weis/Borgwardt rows are shown separately: exact upstream parity was verified only for the official LR subset, while other Weis-format rows test audit portability rather than serve as pooled benchmarks. Asterisks mark low matched support; the second line reports matched retention and valid-stratum count.

prediction rows in DRIAMS-D export; 'not run' denotes no corresponding Weis/Borgwardt LR audit row for that target/site.